

DEVELOPMENT OF THERMOELECTRIC HEAT PUMPING SOLAR AIR HEATER

**A project proposal submitted to the Department of Mechanical
Engineering, University Of Nigeria, Nsukka in partial fulfillment of
the requirements for the award of the Degree of Bachelor of
Engineering in Mechanical Engineering**

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FEB, 2024

CERTIFICATION

I hereby declare that, to the best of my knowledge, DEVELOPMENT OF THERMOELECTRIC HEAT PUMPING SOLAR AIR HEATER is an original work submitted in partial fulfillment for the award of the degree of Bachelor of Engineering in Mechanical Engineering by the University of Nigeria, Nsukka. I certify that it has not been submitted in part or full for the award of the certificate of any other institution or this university.

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APPROVAL

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DEDICATION

I dedicate this project to my family, friends, and myself

ABSTRACT

Heat is transferred from the absorber plate to the air stream by natural convection in conventional solar air heaters (SAHs). This reduces their effectiveness, especially when operating at lower temperatures. To improve heat transfer, this research investigates the creation of a revolutionary SAH design that incorporates thermoelectric (TE) heat pumps.

The idea is to fasten TE modules to the absorber plate's rear. These modules actively pump heat from the absorber plate to the air stream by generating a temperature differential through the application of an electrical current. The design, construction, and performance testing of a prototype TE heat-pumping SAH are the subjects of this study. The impact of operating parameters on the system's thermal performance is discussed in the study, including air mass flow rate, solar irradiation, and electrical power input. The efficacy of the TE heat pumping technique in enhancing heat transfer efficiency is assessed by contrasting the obtained results with those of a traditional SAH. The results aid in the creation of solar air heating systems with higher efficiency for space heating applications.

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Chapter 1

Introduction

1.1 Background of Study

Over 3.2 million quads of the billion quads of energy that are released from the sun every three seconds reach the earth, demonstrating the reliability and endless supply of solar energy. 70% of this energy is absorbed by land, sea, and clouds (a quad is a unit of energy; one quad is equivalent to $2.93 \times 10^{11} \text{ kwh}$) chgbo A. mgbemene Ph.D (mech eng).

Only a tiny percentage of the sun's energy can be captured and transformed into many types of energy, such as thermal energy, chemical fuels, electrical energy, etc. When converted to thermal energy, it can also be used to solve basic household-related energy needs like space heating, crop drying, and any other product or item that needs to be dried. this solar energy can be collected by a system known as a SOLAR AIR HEATER to solve the aforementioned problems. This system comprises various designs that have been researched, which aim at improving its efficiency like thermal energy storage, single pass Air duct, and double pass Air duct. It is built more like a rectangular duct, which comprises four main components, **the absorber plate**(an aluminum sheet), **the casing**, **a glass sheet**, and **a fan**. The system has a flat top, with a glass sheet covering the entire top of the rectangular duct. Solar radiation refracts on this system, passing through the glass sheet and getting absorbed by the absorber plate, (the glass sheet aids in minimizing convection heat loss from the absorber plate to the environment) after which the absorber plate transfers the heat it absorbed to the working fluid (air) by convection in the system's duct below the absorber. A fan extracts the hot air and directs it through the duct to the intended location. To prevent heat losses from the system to the outside, the duct is adequately insulated. Air enters the system from one end and exits from the other end of the duct, after being heated by solar radiation.

Many configurations have been built to enhance heat transfer from the absorber plate to the working fluid, configurations like attaching **fins** to the absorber plate, embedding

phase change materials on the absorber plate, using a **corrugated surface absorber plate**, embedding a **packed bed of thermal storage materials**, and use of **extended surfaces**. This research work is based on embedding a TEC on the side of the absorber plate facing the air duct. prototype of a solar air heater will be illustrated in the picture below.

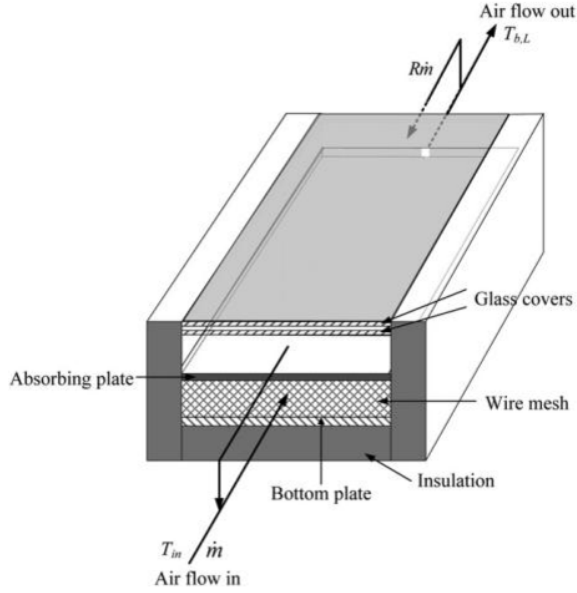


Figure 1.1: SOLAR AIR HEATER Dhatkar et al. (2015)

The remaining chapters will give an explicit discussion of this project.

1.2 Statement of Problem

The effectiveness of heat transfer from the absorber plate to the working fluid determines how well a solar air heater performs. A minute portion of the heat absorbed by the system is lost to the environment through radiation. Because air, the working fluid, is a poor heat conductor and does not absorb and transmit heat at a fast rate. Also, some of the configurations that have been researched do not transfer heat effectively. The major problem every Solar Air Heater has is the ability to transfer heat to the working fluid effectively. Nevertheless, this configuration aims at solving the problem and increasing the efficiency of the system by the use of TECs.

1.3 Aim and Objectives of Study

This project aims to increase the heat transfer to the working fluid and the thermal efficiency of the system. To achieve this goal, the following objectives will be undertaken:

1. Develop a detailed, comprehensive, and conceptual design of a SOLAR AIR HEATER.
2. Construct two prototype SOLAR AIR HEATER with and without THERMO-ELECTRIC COOLERS.
3. Carry out outdoor experimentation on the performance of the SOLAR AIR HEATERS.
4. Analyze and compare test results for the two prototypes SOLAR AIR HEATERS.
5. Provide detailed documentation of the entire process.

1.4 Significance of Study

In solar air heating applications, thermoelectric coolers are a relatively recent technology. There has been relatively little to no research done on the use of thermoelectric modules. Various researchers have tried other approaches to improve the performance of SOLAR AIR HEATERS, such as using different materials for the absorber plates, experimenting with different designs and air flow rates, and attaching different kinds of heat sinks to the absorber. This study is significant because it explores a novel strategy for solar air heating applications.

1.5 Scope of Study

The scope of this project comprises the following:

- Design and fabrication of two prototype SAHs using thermoelectric coolers (TECs) as heat sinks.
- Outdoor testing of the SAHs inlet temperature, outlet temperature, solar intensity, etc.
- Tests will be carried out under clear weather conditions between 10:30 AM to 3:30 PM for 5 days. The readings will be taken at 15-minute intervals each day.
- Graphs of the recorded data will be plotted and the thermal efficiency of each system computed.
- The thermal performance of the two systems will be analyzed to determine the improvement offered by the use of TECs

Chapter 2

Literature review

2.1 Introduction

The need for green energy to replace fossil fuels as the primary source of energy for electric power generation, basic energy needs, and industrial activities (such as manufacturing and heating in industrial processes) is growing daily due to the damaging effects that fossil fuel-generated gasses (CO , CO_2 , NO , and O_3) have on the environment. These gasses cause water pollution, air pollution, global warming, health effects, and other environmental damage. Resorting fully to other forms of energy (renewable energy sources) like **solar energy, thermoelectric energy, wind energy, hydroelectric** etc. that are harmless to humanity and the environment will be a positive step towards energy sustainability and the elimination of global warming. Amongst all these energy sources, solar energy in itself generates more amount of energy than any other renewable energy source ever known, Ogbomo et al. (2017) mentioned in his article that the annual electrical power consumption of the entire planet Earth can be generated by the sun in just an hour, he also stated that solar energy is abundant in nature, in addition to being clean, sustainable and renewable. Kalu-Uka et al. (2022) Solar energy emits energy at a rate of 3.8×10^{23} kW, and is the most plentiful renewable energy source, as is well known. The earth absorbs 1.8×10^{24} kW of this total. The planet could easily use up all of this energy and yet have plenty left over. That is how abundant it is. Nonetheless, this energy source is currently underutilized as a result of the comparatively delayed development and uptake of solar energy-harnessing technology. The simplest and most often used way is to convert solar energy into thermal energy. The vast amounts of energy that the sun freely provides have led many people to view solar energy as a serious source of energy for many years, as stated in k Nikhil Chandra N sai pavan (2022). The easiest and most effective way to use solar energy is to convert it into thermal energy for heating purposes. Solar collectors are the most crucial and fundamental parts of the system needed to convert solar energy into thermal energy.

which are available in several forms, such as solar air heaters (SAH) and photovoltaic modules, but the research scope of this study is based on the former.

Using freely accessible solar energy, solar air heaters generate hot air for industrial or farmer-level drying applications without the need for traditional fuels like electricity, diesel, LPG, firewood, coal, etc. To reduce fuel usage, however, it might be combined with already-existing traditional drying systems including bin driers, conveyor driers, tray driers, and tunnel driers that run on conventional fuels. Because of the low heat transfer capability between the absorber plate and the fluid flowing through the duct, solar air heaters have lower thermal efficiency. To improve the efficiency of solar energy utilization systems like solar air heaters, thermal efficiency needs to be improved by enhancing the heat transfer rate.

Saxena et al. (2015) Different SAH types have been created and tested for specific rating metrics in previous times. In comparison to liquid heaters, solar air heaters offer several appealing advantages in terms of preventing corrosion, boiling, freezing, salt buildup, and leakage. As a result, air heaters are being used in more residential, commercial, and agricultural settings. Providing hot air for the drying of agricultural, textile, and marine products as well as for heating buildings to maintain a comfortable environment, particularly during the winter, is one of the most promising uses for solar air heaters.

There are numerous design options for SAHs when considering materials and combinations, different configurations of solar air heaters will be enumerated and explained in the subsequent section.

2.2 Designs and configurations of solar air heaters

Throughout their study on solar air heaters, several researchers have developed various designs and combinations that aim to maximize the passage of heat from the absorber to the working fluid.

2.2.1 Single Pass Solar Air Heaters

Simple in design, single-pass solar air heaters allow air to travel through the collector just once before exiting the device. Its components are a glass sheet, an absorber plate, and a casing which is a single flow channel with a fluid entry at one end below the absorber plate and a fluid exit at the other. Single-pass systems are less efficient than multi-pass systems because the air only has an opportunity to absorb heat from the collector.

Abdullah et al. (2020) employed reflectors hanging on both sides of the rectangular duct to ensure proper reflection of the sun's rays on the glass sheet, as well as aluminum cans firmly attached on the absorber plate, which were functioning as turbulators to create

turbulence in the flow to collect more heat. At a mass flow rate of 0.05 kg/s, the plate produced a maximum efficiency of 73.4% every day. Raam Dheep and Sreekumar (2020) Nine copper absorber plates in all, measuring 2000 mm in length, 1000 mm in breadth, and 0.1 mm in thickness, were used. The plates were stacked lengthwise, with a 2 mm space between each plate. Each absorber also had circular copper fins attached, measuring 2000 mm in length and 2 mm in diameter. The experiment's greatest efficiency was measured to be 56.8%. This demonstrates that a dispersed fin arrangement with a turbulent airflow is more effective and efficient for heat transfer.

2.2.2 Double-pass solar air heaters(SAH)

Before exiting the system, air passes through the absorber twice, once above and once below, to transmit heat. Because air has two opportunities to come into contact with the absorber before it exits the system, this design is almost always more efficient than the one-pass SAH.

Ramadan et al. (2007) used matrix or porous bed absorbers in an experiment on a two-pass solar air heater. The stuff with which he packed The heater was positioned facing south at a tilt angle of 30° concerning the horizontal to optimize sun radiation on the glass sheet throughout the year. The bed consisted of limestone and gravel. His efficiency reached a level of 57.8%.

Alam (2017) Performance. There is a general decrease in the amount of heat lost to the environment as a result of the air passing beneath the absorber plate extracting heat and the air passing above the absorber plate regaining some of the heat extracted earlier. Improved overall performance is the result of relatively reduced heat losses. Double-pass solar air heaters are further classified as circular flow, counter flow, parallel flow, and recycling based on the direction of the working fluid flow, as demonstrated in the examples below.

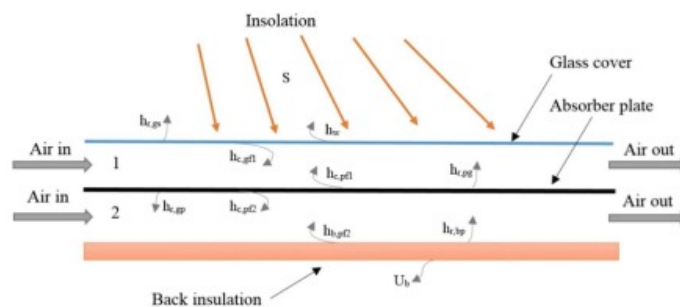


Fig. 2. Parallel-flow, double-pass SAH.

Figure 2.1: Double pass SAH with parallel flow Alam and Kim (2017)

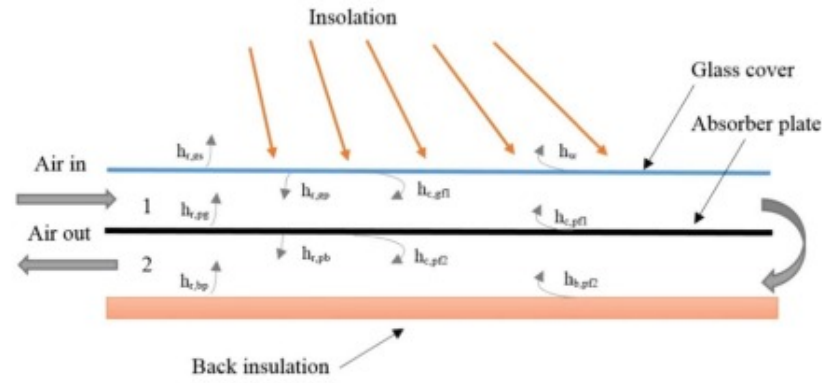


Fig. 3. Counter-flow, double-pass SAH.

Figure 2.2: Double pass SAH with counter flow Alam and Kim (2017)

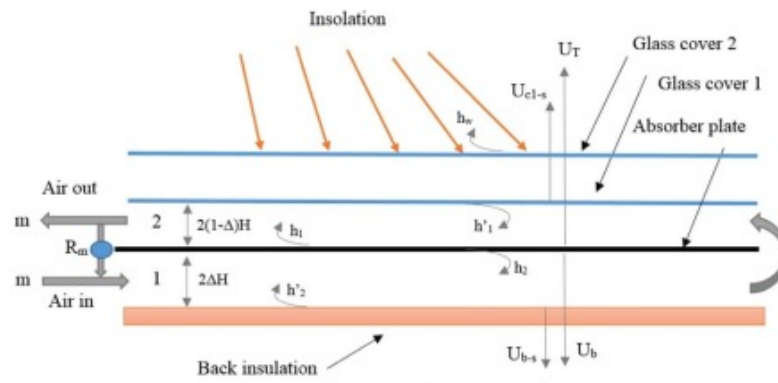


Fig. 4. Double-pass SAH with recycle operation.

Figure 2.3: Recycling or Circular flow double-pass Solar Air Heater Alam and Kim (2017)

2.2.3 Solar air heaters with thermal storage

Thermal storage or phase change material (PCM) solar air heaters are designed to gather and store solar energy for use in air heating at a later time. Paraffin wax and heated mineral oil were used as phase change materials (PCM) in Tyagi et al. (2012)'s experiments on evacuator tube collector-based solar air heaters with and without thermal energy storage (TES).

He discovered that the outlet temperature rose with time, peaked, and then began to decline. Regarding the duration of the experiment, the sun radiation pattern has also been discovered to be similar. The outlet temperature in the non-TES case follows the same pattern, but it has a smaller peak than in the TES examples. Additionally, the optimum temperature rises in all three situations as the mass flow rate increases. Fath (1995) Conventional air heaters are confined to producing hot air exclusively in the presence of sunlight due to the limited heat capacity of the system absorber. Adding a thermal energy storage capacity shouldn't make the already simple conventional heater

design any more intricate. Consequently, by adding an appropriate thermal energy storage material to the tube set, the system's capacity for storing heat will be improved, and because the absorber temperature will drop, the system's heat loss will be decreased, boosting the system's overall efficiency.

Thermal energy can be stored as sensible or latent heat, generally speaking. From the standpoint of energy storage, latent heat thermal energy storage is a suitable and efficient way to store heat. Large thermal capacity and continuous storage are two of its advantages.

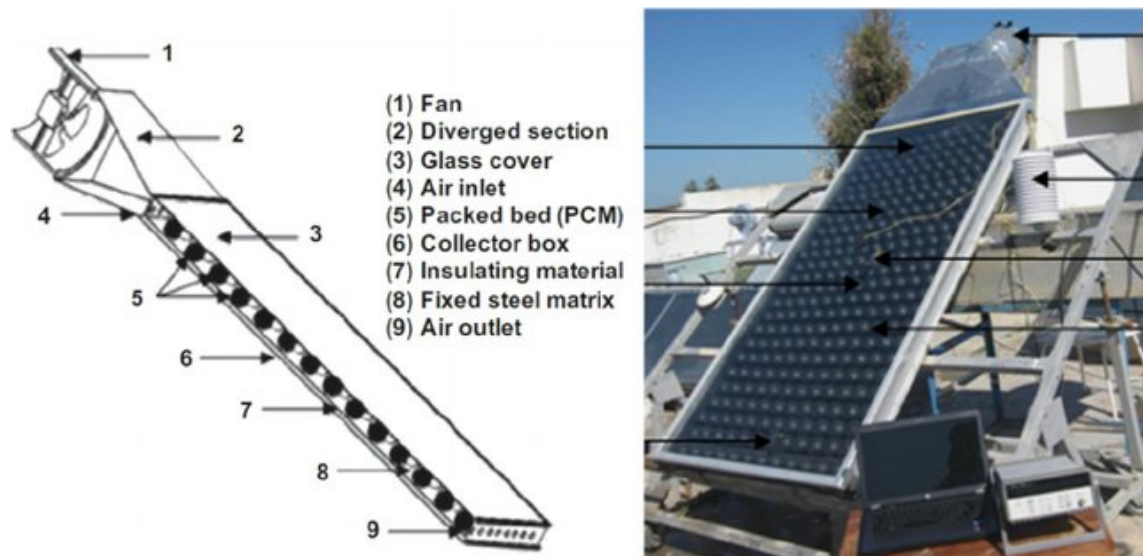


Figure 2.4: Solar air heaters with thermal storage Sharif et al. (2015)

2.3 Factors affecting the performance of solar air heaters

2.3.1 Use of extended surfaces (fins)

Ghritlahre et al. (2022) worked on a solar air heater where the mass flow rate range was $\dot{m} = 0.03 - 0.06$ kg/s and the absorbers were fixed and free fins attached. Their results show that the fixed fin SAC works better than the free fin collector. After the experimental investigation, they found that the collector efficiency also increased with the length of the surface (fins) attached to the flow region. After constructing absorber plates with and without fins, they tested the energy and thermal efficiency for the fin-equipped, fin-free, and varied air flows ($25, 50, \text{ and } 100 \text{ m}^3/\text{m}^2 \text{ h}$) fin-covered plates at $0^\circ, 15^\circ, \text{ and } 30^\circ$ tilt angles. With an airflow velocity of $100 \text{ m}^3/\text{m}^2 \text{ h}$ and a tilt angle of 30° , Type II reached a maximum energy efficiency of 39.05%. On the other hand, Type-II showed a maximum energy efficiency of 0.83% at $100 \text{ m}^3/\text{m}^2 \text{ h}$ and a tilt angle

of 0° . Additionally, they claimed that due to the larger temperature differential, the finned SAH with a double glass cover was more effective than SAH without fins.

El-Bialy and Shalaby (2023) Using aluminum cans as extended surfaces examined how employing exterior reflectors and aluminum cans affected a single-pass flat-plate SAH's thermal efficiency. There were three different absorber plate arrangements. The initial one is a flat-plate method devoid of stationary cans. the second and third ones are both flat-plate absorbers with cans placed on the top surface in aligned and staggered arrangements, respectively. The types 2 and 3 aluminum cans were opened on both sides. Using the fixed cans improves the heater's thermal performance by expanding its heat transfer area and intensifying the airflow's turbulence. The daily efficiency of the staggered and aligned cans flat plate SAH was improved by approximately 32% and 28%, respectively, by using bottom and top external reflecting mirrors in comparison to the flat plate SAH with a mass flow rate of 0.05 kg/s of air. a pictorial representation of the aluminum cans on the flat plate is shown below.



Figure 2.5: Aligned and staggered absorber El-Bialy and Shalaby (2023)

2.3.2 Use of packed bed absorber materials

It has been stated that a SAH's air duct's thermal efficiency can be increased by using a porous or packed bed medium. Saxena et al. (2015) stated that, it has been noted that solar thermal devices function better by when a high-quality material for energy storage is provided. High-quality Thermal energy storage(TES) materials extend a solar device's ability to store heat for extended periods. Subterranean aquifers, "designed containers,"



Figure 2.6: A complete SAH with aluminum cans El-Bialy and Shalaby (2023)

”soils,” ”lakes, bricks, and ingots” are some examples of these energy storage devices. Thermal energy is stored in these systems as both latent and sensible heat.

As a result, working throughout the day, in inclement weather, or at night becomes more difficult. Therefore, it is thought that one major issue impeding the advancement and promotion of solar energy is the absence of reliable heat storage devices. An attempt has been made to create a SAH that can function effectively all year long. The current designs differ significantly from the earlier ones, either because they use thermal heat storage or hybrid SAHS. Goel et al. (2021) studied a Solar Air heater system working with thermal storage with the materials being able to conduct heat of high temperatures of about 150°C to 1100°C , these materials were used in a double pass solar air heater whereby the bottom was provided bottom extending longitudinal fins. the thermal efficiency was 69%.

2.3.3 Use of multi-pass airflow channel

The use of multi-pass airflow channels in solar air heaters(SAH) greatly improves the performance of the system, this is because Air makes a double pass in the system(below and above the absorber) thereby picking enough heat before leaving the system ”it can be said that the Air does a two-factor authentication before leaving the system”. this ensures that the working fluid is thermally saturated for applications.

Ravi and Saini (2016) Studied different configurations of multi-pass solar Air heaters and found out that Under similar operating conditions, the thermal performance of the double-pass solar Air heaters (DPSAHs) was found to be better in comparison with conventional single-pass solar air heaters (SPSAHs). From an economic point of view, the DPSAH with a single glass cover was the most cost-effective. In smooth DPSAH, the maximum thermal efficiency was obtained if the depths and the MFR in both flow

channels were kept the same.

Aldabbagh et al. (2010) Worked on a double-pass solar air heater with two glaze sheets with a distance of 5cm between them and the second glass has a distance of 10cm from the bottom of the collector. In his project, he used 10 steel wire mesh with a cross-section of 0.2×0.2 cm painted black, as a porous bed material. the wire mesh was arranged in such a way that it had a distance of 1cm from each one, up until the last wire. after the solar intensity has entered the system through the glazing, it goes through the wires and at this point the working fluid also flows through the wires absorbing the heat instantaneously, so before the insolation gets to the last wire, it has almost been completely absorbed. thus, in this system, the wires are the absorber.

After the experiment, the efficiency obtained was 83.5% as at 13:00 h, at the peak of irradiance of the sun, with a mass flow rate of 0.038 kg/s and a temperature difference of 38.8°C . he also worked on a single pass solar Air, as at noon, the efficiency he got was 45.93% with the same mass flow rate and a temperature difference of 27°C . The above experimentation and analysis have shown obviously that a double-pass solar air heater is very much more effective than a single-pass solar Air heater.

2.3.4 Thermoelectrics

To convert heat energy into electrical energy or electrical energy to a temperature gradient, a thermoelectric device is a special kind of solid-state energy converter. It works on a circuit that combines thermal and electrical effects. It is very dependable, ecologically friendly, and devoid of moving parts. The thermoelectric generator (TEG) and the thermoelectric cooler (TEC) are the two types of thermoelectric modules.

An assortment of n-type and p-type semiconductor thermoelements coupled thermally in parallel and electrically in series make up the thermoelectric module. The most often used thermoelement is Bismuth Telluride Bi_2Te_3 . The Bismuth Telluride bars are electrically insulated and have two ceramic insulators acting as a foundation between them. They are linked at the ends. A cutaway diagram of a thermoelectric module operating in power-generating mode is shown in Figure 2.1. chgbo A. mgbemene Ph.D (mech eng)

The thermoelectric module (TEM) can function as a heating or cooling device based on the availability of electricity. Figure 2.8 depicts how the TEM is arranged. Figure 2.8a illustrates how electricity is supplied to the TEM, which causes heat to be absorbed from one side and dissipated to the other, causing the sides to heat or cool. This is made possible by a phenomenon called the Peltier effect, in which a TEM transforms electrical energy into a temperature gradient. Enescu and Virjoghe (2014). When operating as a TEG, it can also be utilized to produce electricity due to a phenomenon known as the Seebeck effect. This effect operates by causing a charge carrier (N-type) to gravitate

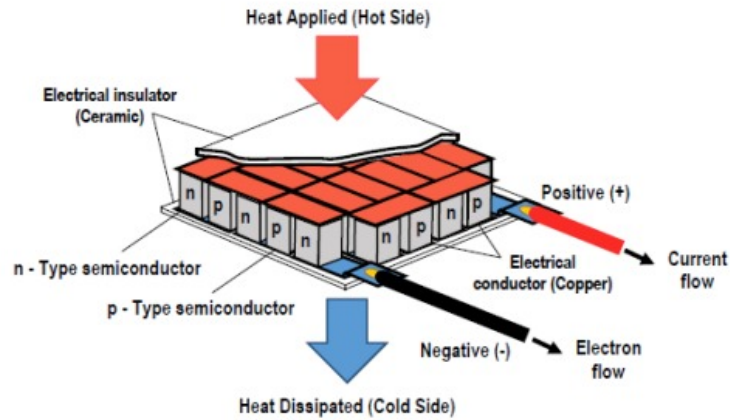


Figure 2.7: A cutaway diagram of a thermoelectric module in power generation mode
 chgbo A. mgbemene Ph.D (mech eng)

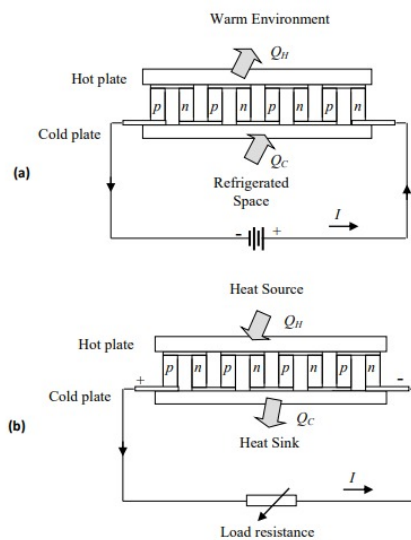


Figure 2.8: Modes of thermoelectric module usage (a) as a cooling/heating device, (b) as a power generator.
 chgbo A. mgbemene Ph.D (mech eng)

from a hot junction to a cold junction (P-type) when a semiconductor (SE) is heated. The uneven dispersion of charge carriers contributes to the accumulation of charge that generates back electromotive force and inhibits additional charge transfer. Thomas J. Seebeck made the initial observation of this heat energy conversion event in 1821. The TEG junctions consist of different p-type and n-type materials. Both thermally and electrically, they are linked in parallel. More holes are present in p-type junctions than in n-type junctions, while more electrons are present in n-type junctions. Mal et al. (2017) The figure of merit (Z), which indicates the thermoelectric material's efficiency, is one of the module's features that has a significant impact on the module's total performance. The dimensionless form " ZT ," where T represents the absolute temperature, is frequently used to express it. The thermoelectric materials' thermal conductivity (λ),

Seebeck coefficient (α), and electrical conductivity (σ) all have a direct impact on the figure of merit "Z". $Z = \frac{\sigma^2 \alpha}{\lambda}$ is one way to express the figure of merit Z Gao et al. (2016). Numerous benefits come with thermoelectric devices, including *direct energy conversion, no internal moving parts or working fluid, a long lifespan, no scale effect, noiseless operation, and suitability for any working position*. Champier (2017)

Numerous applications, including the production of electricity, refrigeration and cooling, and heat pumping, have shown the value of thermoelectric devices. Low electrical resistivity, low thermal conductivity, and a high Seebeck coefficient are desirable properties for a thermoelectric material Jaziri et al. (2020)

More details of its application will be discussed below.

2.4 Applications of Thermo electrics to Solar Devices

The application of thermoelectrics to solar energy conversion devices includes:

2.4.1 refrigeration and cooling

Important uses for thermoelectric modules include serving as refrigerators in remote, off-grid rural communities. Xi et al. (2007) created a thermoelectric refrigerator powered by solar energy. The side of this refrigerator has embedded TECs that are powered by PV modules. The system functions flawlessly during the day when there is strong insolation, but at night, when the sun must have set, considerations were made and a battery was added to the system to make up for the period or days of autonomy (when the cloud has no sun). the system's effectiveness It was discovered that η_{system} was less than 0.06.

Hara et al. (1998) A thermoelectric helmet was created to provide comfort for individuals when walking, cycling, and sitting. In this setup, the forehead was cooled by the cold side using a thermoelectric element placed beneath the brim's root. The thermoelectric element was powered by solar cells attached to the headgear's brim and top. To act as a heat sink, a sizable aluminum fin, and an axial cooling fan were fastened to the hot side of thermoelectric elements. Three models were created, each serving a distinct function. It was reported that sixteen pieces of silicon crystal solar cells ($0.6V_{max} \times 0.4A_{max}$), each measuring 20×60 mm and weighing 2g, and a thermoelectric module ($3A_{max} \times 14V_{max}$) measuring 30×30 mm could provide a maximum temperature difference of $4-5^\circ C$ for thermal comfort. Three pieces of amorphous flexible paper-type solar cells ($0.35V_{max} \times 1.5A_{max}$), each measuring 90×230 mm and weighing 5g, combined with a 40×40 mm thermoelectric module ($4.0A_{max} \times 17.5V_{max}$) and a lighter electric fan allowed for the creation of headgear with a minimum weight of 135g.

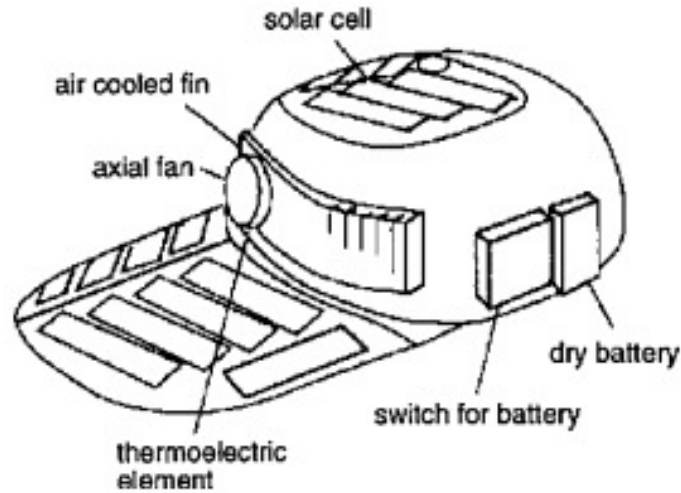


Figure 2.9: Thermoelectric headgear Xi et al. (2007)

2.4.2 Electricity generation

PV arrays can be utilized to transform solar thermal energy into temperature differential, which thermoelectric devices can then use to operate as a heater or cooler. Additionally, these gadgets can transform solar energy into electrical energy for use as power generators. When Tyagi et al. (2023) investigated PV-TEG hybrid generators and solar thermoelectric generators (STEGs), he discovered that STEGs use sunlight to create a temperature gradient that aids in the production of power through the Seebeck effect. The temperature gradient in a TEG can be raised by combining a concentrator and thermoelectric generator. According to Rockendorf et al. (1999), a solar thermoelectric generator (STEG) converts waste heat that the panel is unable to use into additional electrical energy by using thermoelectric generators (TEGs). The STEGs can enhance and optimize the system's overall efficiency. The amount of power generated inside the TEG is determined by the temperature differential that arises there. On the other hand, he said that heat is first produced in STEGs and subsequently transferred to the thermal resistance of the TEG, producing electricity. On the other hand, a PV-TEG hybrid generator generates heat and electricity simultaneously by using photovoltaic cells that are laminated on the surface of a solar absorber and cooled by a liquid heat-transfer medium.

2.4.3 heating

In countries near the South Pole, thermoelectric modules are utilized for water heating and space heating in the winter. Using a photovoltaic module to power the TECs, Liu et al. (2015) created a method that maximizes the heat from the TECs when current is provided to them to heat spaces and heat water. A water tank is filled with water when

the system starts up, and as return air enters the interior through heat exchangers, it cools down. Concurrently, the heat exchanger situated across from the TEC modules heats the water in the water tank. Consequently, thermal energy from the heated surfaces of TEC modules can be extracted. while the room is heated The heat from the thermoelectric modules' cold side is dispersed by the flow of water during the winter when the direction of current input is reversed and the heating and cooling sides alternate. After that, the system can function as a thermoelectric heat pump for water sources. When air enters an interior space for space heating, it is heated using heat pipe exchangers; this system has a coefficient of performance (COP) of 3.01.

2.5 Justification for study

Solar air heaters are a vital component of the energy industry, and they are used in many different industries where hot air is needed for processes like desalination, drying, and heating. It uses a renewable energy source, which is more dependable than conventional energy sources like fossil fuels, which are harmful to the environment. This has led to an increase in research on how to enhance the efficiency of solar air heaters, to which this study belongs. Because thermoelectric modules are environmentally friendly and can massively transfer heat from a place of higher temperature to a location of lower temperature, using them in conjunction with solar air heaters will be advantageous.

Chapter 3

Methodology

The materials and components used for the construction of the solar air heaters are listed below.

1. Casing: The other SAH components are housed in the shell that is seen below. It is constructed from 20mm-thick plywood. The dimensions of the case are 1500 mm in length, 750 mm in width, and 150 mm in height. A pair of woods, measuring 25 x 20 mm in cross-section, are fastened to the side and end walls, with a 2 mm gap between them. While the absorber is held in between the woods, the glass is positioned on top of the first wood and securely secured in place with gum. The front of the shell features a straight opening for cool air to enter. The other end has 5 holes drilled with DC fans installed which blow hot air out of the system. The same casing design will be used for the two configurations. A CAD design of the casing is represented below:

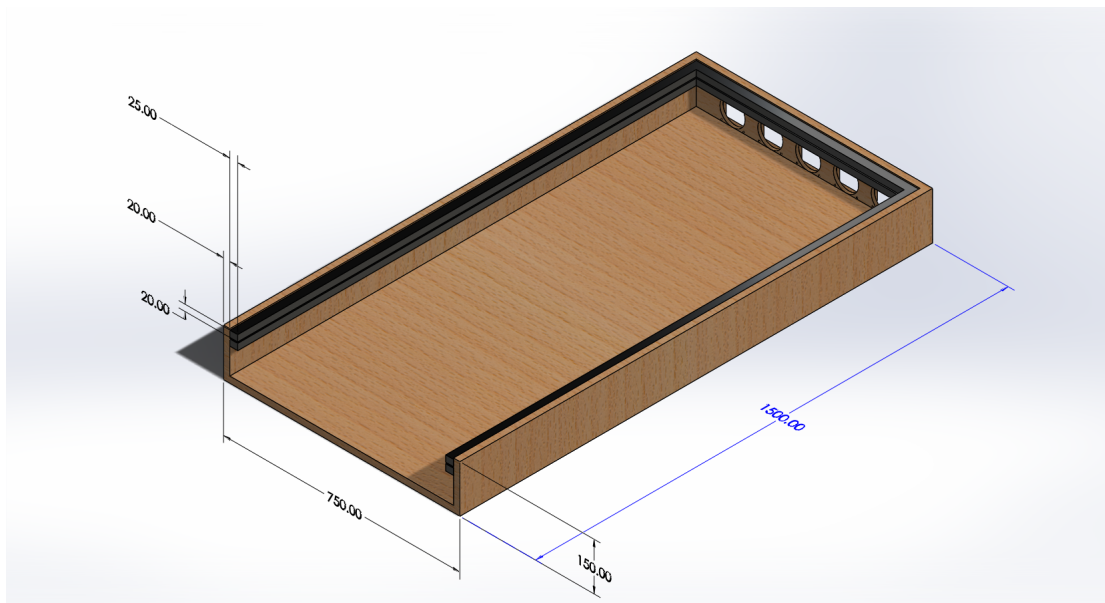


Figure 3.1: Wooden casing

2. Glazing: The glazing is a transparent glass cover having a thickness of 4mm. Solar insolation enters the device through the glazing. The glass sheet rests on the top rectangular wood inside the casing. A CAD design of it is shown below:

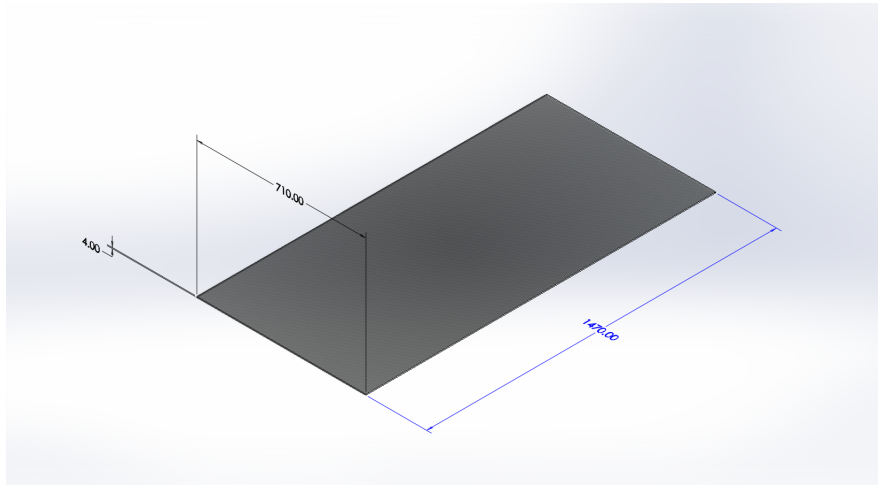


Figure 3.2: A transparent glass

3. Absorber: The absorber plate is an aluminum sheet of 2 mm thickness. For the conventional SAH, the absorber plate is just the aluminum sheet while for the improved SAH, the absorber plate has 11 TECs attached to it. The absorber plate for each configuration will be placed in between the two woods attached to the sides and ends of the casing. The distance between the glass and absorber plate is 20mm.

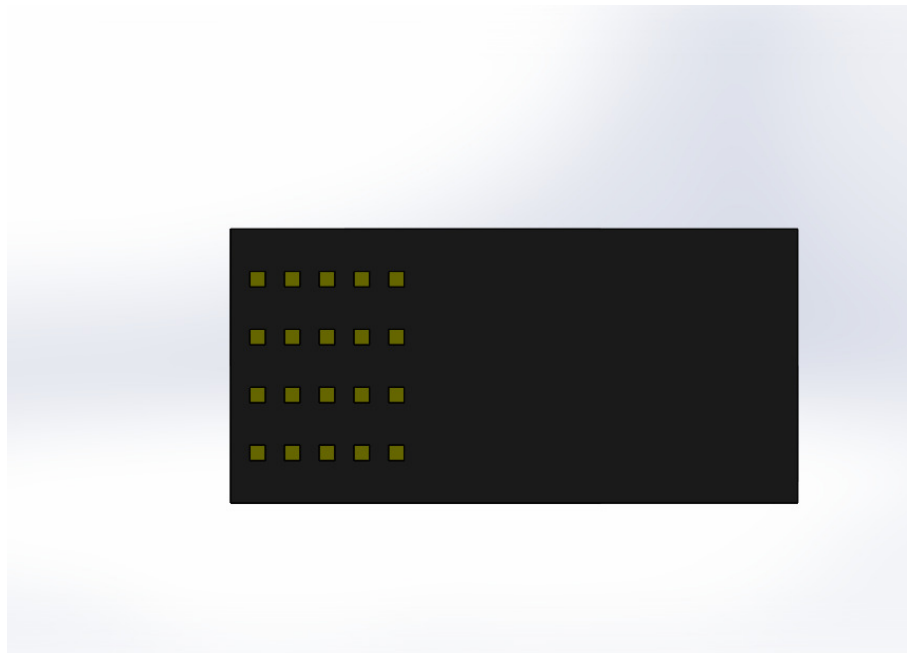


Figure 3.3: Absorber plate with 11 TECs attached to it

4. Insulation: The sides and the bottom of the collector are insulated with a 25 mm thick glass wool to minimize heat loss by conduction. The insulation is screwed to the walls of the solar collector.
5. DC Fans: 5 12V DC fans are attached to the back of the solar collector or casing for the two configurations. These fans are intended to suck the hot air inside the collector out of the system.

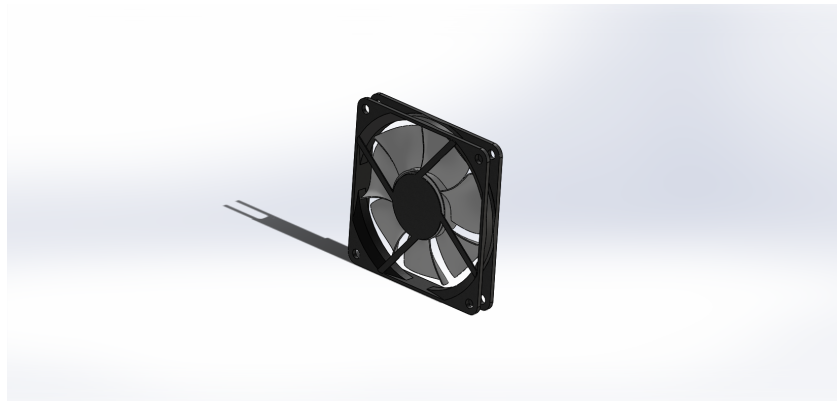


Figure 3.4: A 12 Volt DC fan

6. TECs: 11 Thermoelectric coolers are attached to one-third portion of the absorber plate with a horizontal distance of 85mm and a vertical distance of 82.5mm in between them.. Each TEC has a dimension of 40 mm $\times$ 40 mm with a voltage and current rating of 6V and 5A respectively.



Figure 3.5: Thermoelectric Cooler

7. Photo Voltaic Module: 4 PV modules are used to power the DC components (fans and TECs). Each module has a power rating of 40 watts.

3.1 Instrumentation

1. K-Type Thermocouples: A multi-tester thermocouple will be used to measure the temperature at different points on the absorber plate. It would be placed at six different positions on the solar air heater to measure the temperatures at these locations.
2. Solar meter: It is used to measure the amount of solar insolation (solar intensity) during testing and/or experimentation.
3. Anemometer: This is used to measure wind speed to determine the mass flow rate of air in and out of the solar collector.

3.2 Cost Analysis

The prices and quantities of materials and components needed for the construction of the two solar air heaters are presented below:

Table 3.1: Cost analysis for the two configurations of SAH

Item	Quantity	Unit price	Total
Transparent glass (4mm thickness)	2	₦15,000	₦30,000
Aluminium sheet (2mm thickness)	2	₦31,500	₦63,000
3/4" MBF board	2	₦26,000	₦52,000
Heat sinks and shipping	20	₦1,700	₦34,000
DC fans	10	₦1,000	₦10,000
TECs and shipping	11	₦3,100	₦34,100
Screws (a packet)	-	₦3,000	₦3,000
Thermal paste	1	₦2,000	₦2,000
Edge tape gum (7 ft)	-	-	₦1750
Cellicone glue	1	₦2000	₦2,000
Spray paint	2	₦2,000	₦4,000
Oil paint	1	₦1500	₦1500
PVC wires	11 yards	₦910	₦10,000
Workmanship	-	-	₦28,000
Transportation	-	-	₦25,000
Total	-	-	₦300,350

3.2.1 Experimental procedure

two different types of SAH configurations were tested under the same outdoor conditions in the wet season of 2024. Tests were carried out for 3days on April 12, April 16, and April 17, 2024. The readings were taken at intervals of 15 minutes. an inclination angle of 14_o was used for both collectors considering the latitude of the University of Nigeria, Nsukka. For maximum incident solar radiation on the collectors. The tests span from 10:30 AM to 3:30 PM for each of the testing days. four PV modules were used to power the DC components, three modules were connected to the TECs, and one module was used to power the fans. The PV modules were connected directly to the components. This means the voltage, current, and mass flow rate of air in the system varies as solar insolation rises and falls throughout the experiment. an anemometer was used to measure the speed of air coming out from the conventional SAH, V_{e1} , and the SAH with TECs, V_{e2} . A multimeter was used to measure the supply voltage and cur-

rent across the fans in the TE-SAH V_2 , I_2 and across the fans in the conventional SAH V_3 , I_3 . The multimeter also measures the supply voltage and current across the TECs V_4 , I_4 . A digital thermometer was used to measure the ambient temperature T_1 , outlet temperature from the conventional SAH T_2 , and outlet temperature from the SAH with TECs T_3 . A solar meter was used to measure solar insolation at different times of the day. A picture of the setup and that of the wiring diagram is shown below.

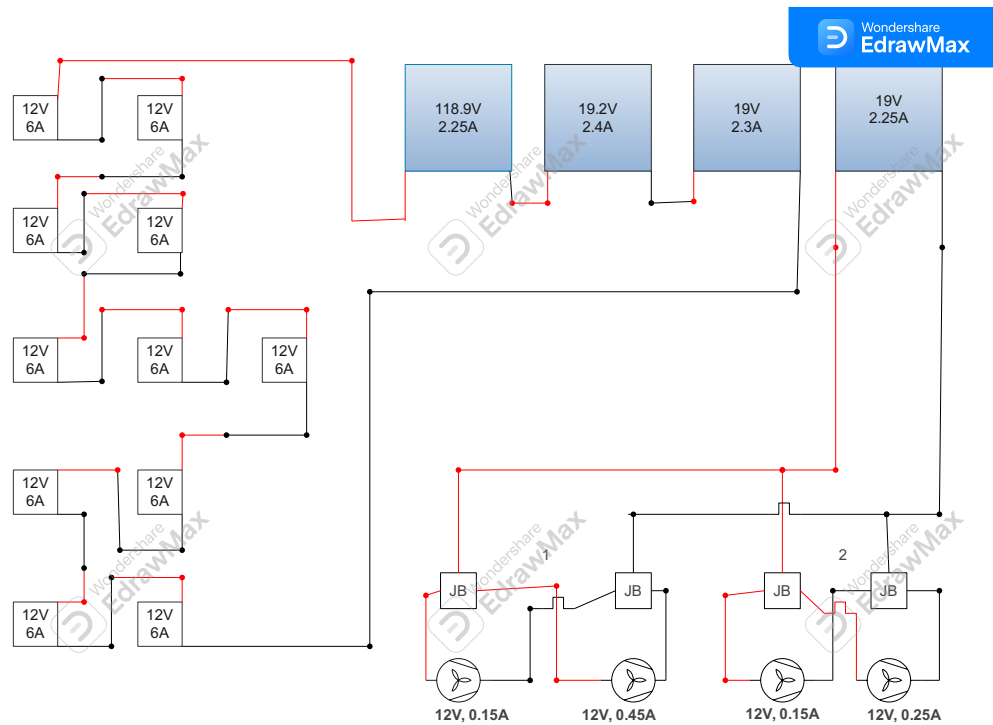


Figure 3.6: Picture of the wiring diagram, 2 are the fans without TEC and 1 are the fans with TEC



Figure 3.7: Picture of the experimental setup

Table 3.2: Design Specifications of each solar air heater

Parameter	Value
Dimension of casing	$1.5m \times 0.75m \times 0.16m$
Length of glass	1.47m
Width of glass	0.72m
Glass thickness	4mm
Area of collector	$1.47 \times 0.72 = 1.0584m^2$
Air duct height	8.5cm
Distance between glass and absorber plate	3cm
Number of fans	2
Internal diameter of fan	8cm
External diameter of fan	3.8cm
Effective area of exit hole	$\frac{1}{3} \cdot \frac{\pi}{4} (0.08^2 - 0.038^2) = 0.0013m^2$

3.2.2 Analysis of thermal collector performance

Instantaneous and overall thermal efficiency

The thermal efficiency of the collector is determined by the ratio of useful heat gain $\dot{Q}_u$ to the solar insolation $\dot{Q}_s$. The useful heat gain is expressed thus: $\dot{Q}_u = \dot{m} \times c_p \times (T_{out} - T_{in})$

where $\dot{m}$ is the mass flow rate of the air in (Kg/s), c_p is the specific heat capacity of air (KJ/Kg · K), T_{out} is the outlet temperature of air ($^{\circ}C$), and T_{in} is the inlet temperature ($^{\circ}C$). For this experiment, the specific heat of air is $c_p = 1.005 \text{ KJ/KgK}$. The mass flow rate can be calculated thus:

$$\dot{m} = A_e \times V_e \times \rho$$

Where V_e is the velocity of air at the exit (m/s), A_e is the exit area (m^2), and $\rho = 1.127 \text{ kg/m}^3$ is the density of air. The insolation from the sun is utilized by the collector and the DC components. Therefore the power input to the system is:

$$\dot{Q}_s = I_s \times A_{collector} + \sum IV \quad (3.1)$$

where I_s is the solar intensity, V is the voltage and I is the current across the DC components. Therefore, the efficiency of the system at any instant is:

$$\eta_i = \frac{\dot{Q}_u}{\dot{Q}_s} = \frac{\dot{m}c_p(T_{out} - T_{in})}{IA_c + \sum IV} \quad (3.2)$$

For the modified SAH with TECs, the instantaneous efficiency is

$$\eta_1 = \frac{(m_{1i} + m_{1ii})c_p(T_3 - T_1)}{IA_c + I_2V_2 + I_4V_4} \quad (3.3)$$

where m_{1i} and m_{1ii} are the mass flow rates of air coming out from the two holes at the exit. For the conventional SAH, the instantaneous efficiency is

$$\eta_2 = \frac{(m_{2i} + m_{2ii})c_p(T_2 - T_1)}{IA_c + I_3V_3} \quad (3.4)$$

where m_{2i} and m_{2ii} are the mass flow rates of air coming out from the two exit holes.

3.3 Overall heat loss coefficient(U_L) and heat removal factor(F_R)

The solar insolation reaching the collector is not fully absorbed. Some of the radiation is lost to the environment by conduction and reflection from the glazing and walls. The heat loss to the environment is expressed as:

$$Q_l = U_L \cdot A(T_{out} - T_{in}) \quad (3.5)$$

Where U_L is the heat loss coefficient, A is the collector area, T_{in} and T_{out} are the temperatures of air into and out of the system. The useful heat gain of the collector Q_u is given by the difference between the insolation energy and the heat loss energy which is expressed below:

$$Q_u = Q_i - Q_L \quad (3.6)$$

All the variables affecting the heat-exchange efficiency of a flat-plate solar heat collector can be summed up into one "efficiency factor". These efficiency factors are essentially design constants of the particular collector design and are only slightly impacted by operating conditions. The figure that links a collector's actual useful energy gain to the useful energy gain that would happen if the entire collecting surface were at the fluid inlet temperature is known as the heat removal factor (FR), also often called this efficiency factor. It is said as follows:

$$F_R = \frac{\dot{m}c_p(T_{out} - T_{in})}{A[I(\tau\alpha) - U_L(T_{out} - T_{in})]} \quad (3.7)$$

F_R is an important design parameter because it estimates the heat resistance that the absorbed solar radiation must overcome to reach the collecting fluid. The absorber area and shape, flow type, and function all affect the collector efficiency factor η_i , which in turn affects this component. When a solar collector reaches the fluid inlet temperature, it can store the most useful energy possible. Rearranging equation (3.9) and substituting $Q_u = \dot{m}c_p(T_{out} - T_{in})$, the following equation is obtained:

$$Q_u = F_RA[I(\tau\alpha) - U_L(T_{out} - T_{in})] \quad (3.8)$$

Dividing through by AI , the left-hand side becomes the instantaneous efficiency, and equation (3.10) reduces to

$$\eta_i = \frac{F_RA[I(\tau\alpha) - U_L(T_{out} - T_{in})]}{AI} = F_R(\tau\alpha) - F_RU_L \left(\frac{T_{out} - T_{amb}}{I} \right) \quad (3.9)$$

Equation (3.11) is in the form of equation of a straight line

$$y = mx + c \quad (3.10)$$

where $y = \eta_i$, $x = \frac{(T_{out} - T_{in})}{I} = \frac{\Delta T}{I}$, $c = F_R(\tau\alpha)$ is the vertical intercept and $m = -F_RU_L$ is the slope.

The above parameters to be determined at the end of the experiment is essential in the computation and analysis of the performance of the system using graphs. The experiment will also involve an infrared imaging of the absorber plate to obtain a heat map that will show the cooling effect of the TECs. The results will be analyzed to compare the two SAHs and the efficiency improvement offered by the use of TECs. The following graphs will be plotted

- Solar intensity (I) vs time of day
- Inlet and outlet temperatures T_{in} and T_o vs time of day
- Instantaneous efficiency η_i vs time of day
- Instantaneous efficiency η_i vs $\frac{\Delta T}{I}$
- Voltage (V) vs time of day
- Current (A) vs time of day

Chapter 4

Results and Discussion

4.1 Data Collection

The experimental data was collected for 3 days from 10:30 AM to 3:30 PM at 15-minute intervals. The data was collected using the measuring instruments described in Chapter 3. The variables which were measured during the experiment are presented in the table below. Microsoft Excel was employed in the documentation of the experimental data while Matlab was used to analyze and plot the graphs.

Table 4.1: Variables measured during the experiment

Variables	Meanings
I	Solar insolation (W/m^2)
V_2	Supply voltage across fans connected to TE-SAH
I_2	Supply current across fans connected to TE-SAH
V_3	Supply voltage across fans connected to conventional SAH
I_3	Supply current across fans connected to conventional SAH
V_4	Supply voltage across TECs
I_4	Supply current across TECs
T_1	Ambient temperature
T_2	Outlet temperature of air from conventional SAH
T_3	Outlet temperature of air from SAH with TECs

4.2 Performance Analysis I

This section presents an analysis of the data obtained for both types of SAHs under study for the first day of the experiment. Figure 4.1 shows the variation of solar insolation with time of day. As the solar intensity rises and falls, the voltage and current supplied to the components (TECs and fans) also increase and decrease as seen in Fig 4.2 and Fig 4.3. as well as the mass flow rates.

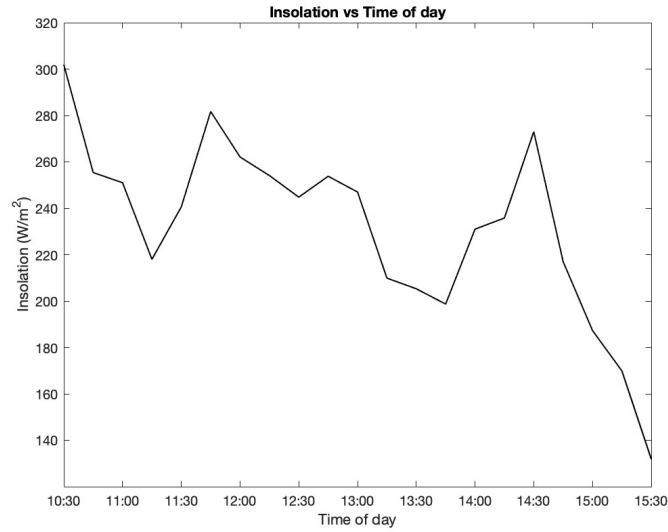


Figure 4.1: Variation of insolation with time of day

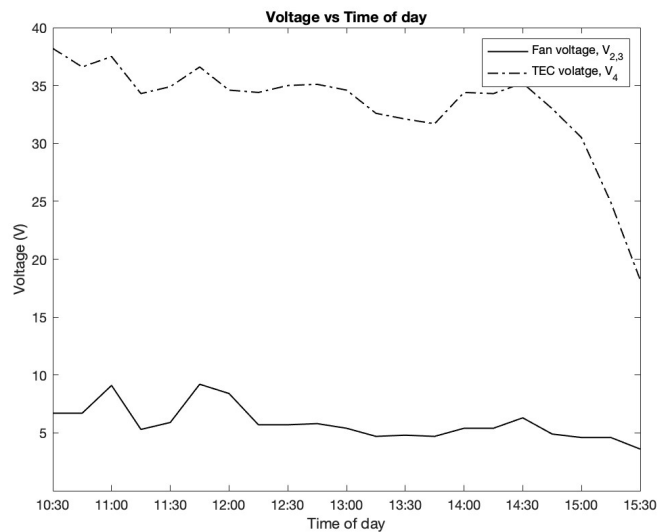


Figure 4.2: Variation of voltage across fans and TECs with time of day

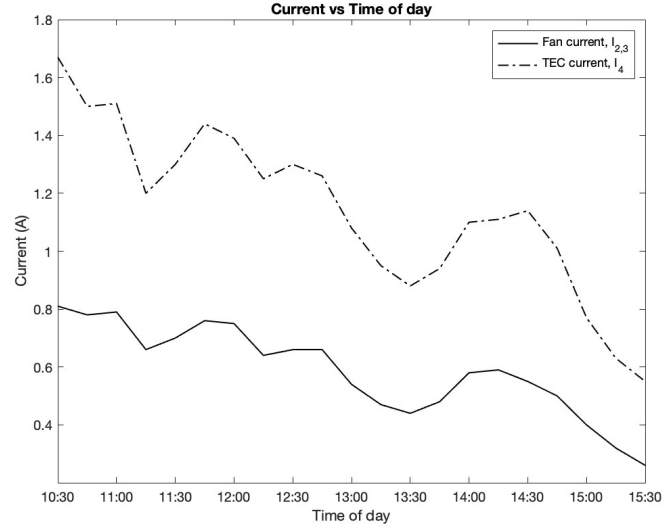


Figure 4.3: Variation of current across fans and TECs with time of day

4.2.1 Temperature and Efficiency Analysis I

The fluctuation in the ambient air temperature at the intake and exit of the two SAHs is depicted in Figure 4.4. It is evident that, at every instant, the air exiting the modified SAH with TECs has a temperature higher than the output temperature of the regular SAH. At 12:15 PM on this particular day, the modified SAH recorded a maximum temperature of 58° , whereas the traditional SAH reported a maximum temperature of 56° . A higher heat gain is typically indicated by the modified SAH's air outlet temperature, which is typically 4° to 6° higher than that of the standard SAH.

Figure 4.5 demonstrates the two SAHs' immediate efficiency at various times of the day. Equations 3.5 and 3.6 are utilized in the computation of the efficiency. As the solar insolation increased, the SAHs' efficiency generally increased between 10:30 AM and 12:00 PM. The modified SAH recorded a maximum efficiency of 28% at 11:15 PM, which represents an efficiency improvement of approximately 33%, compared to the traditional SAH's maximum efficiency of 21% at 12:15 PM. On this particular day, afternoon, insolation began to decline, which also resulted in a decline in efficiency as shown in the graph. This decline continued until evening. The results indicate that the modified SAH with TECs has a better performance than the conventional SAH both in terms of heat gain (outlet temperature) as well as efficiency improvement.

4.3 Performance Analysis II

On the second day of the experiment, the solar insolation vs time graph revealed a rise in insolation throughout the day, peaking at 380.6 W/m^2 at noon. For the day, the average

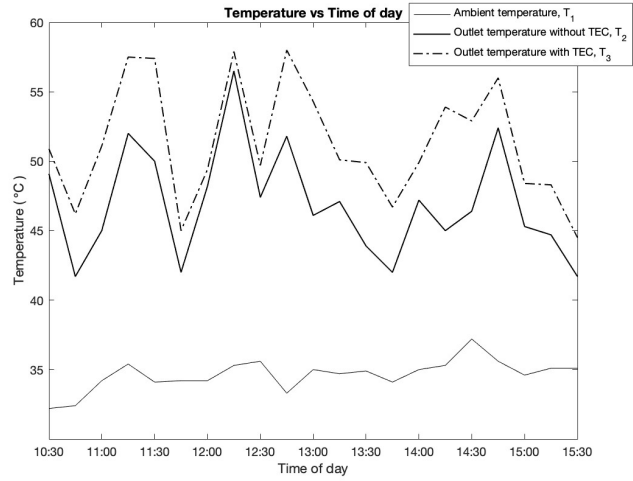


Figure 4.4: Variation of temperature with time of day

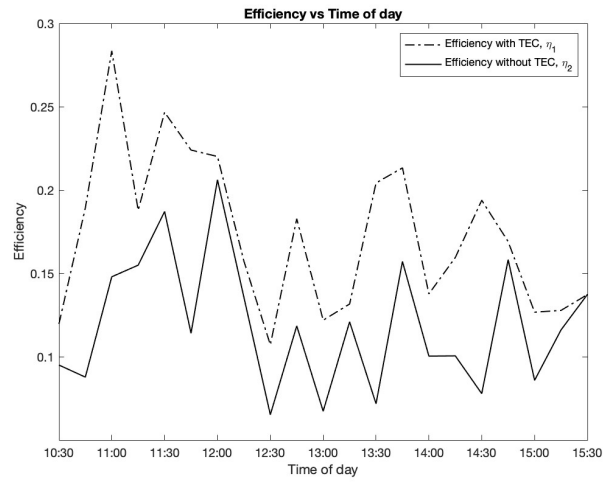


Figure 4.5: Variation of instantaneous efficiency with time of day

insolation was 278 W/m^2 . When the voltage and current between the components rose and fell in response to solar intensity, a similar pattern was seen.

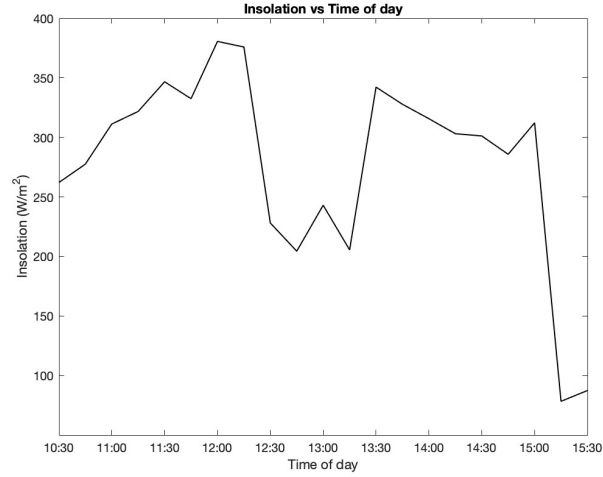


Figure 4.6: Variation of insolation with time of day

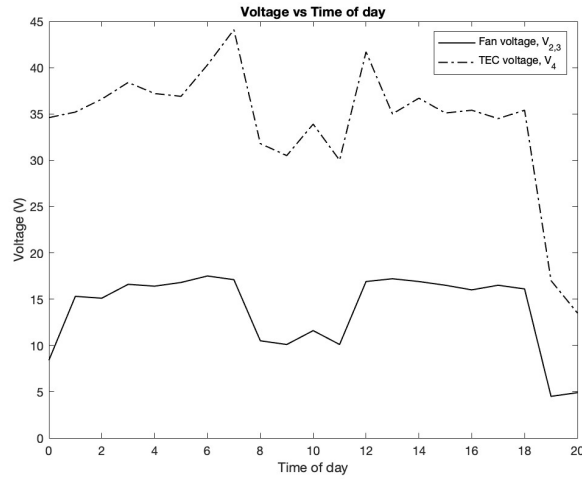


Figure 4.7: Variation of voltage across fans and TECs with time of day

4.3.1 Temperature and Efficiency Analysis II

On the second day of the experiment, Figure 4.9 depicts how the temperature varied for the SAHs according to the time of day. The graph indicates that when the air temperature from the SAHs reached its first high, it climbed from 10:45 AM to 12:00 PM. At this stage, the air outlet temperatures from the modified SAH are 57.5° and 53° for the standard SAH. After that, the temperature begins to decline as solar insolation briefly decreases before rising once more to reach its maximum point of the day at 1:30 PM. The modified SAH's maximum outlet temperature is 61° , while the standard SAH's maximum outlet temperature is 55° . After this, the temperature declines till the conclusion of the day. The results show that the outlet air temperature with TECs was higher than the outlet air temperature without TECs.

Figure 4.10 shows the results of experiments regarding the efficiency variations of the SAHs at different times of the day. It was observed that the efficiencies increase to a maximum value at 1:30 PM, and then they start to decrease in the evening as the time proceeds further. The maximum efficiency recorded for the conventional SAH was 55% while the maximum efficiency for the modified SAH was 72% at 1:30 PM. The efficiency graph as well as the temperature, voltage, and current graphs show a similar trend as the solar insolation throughout the day.

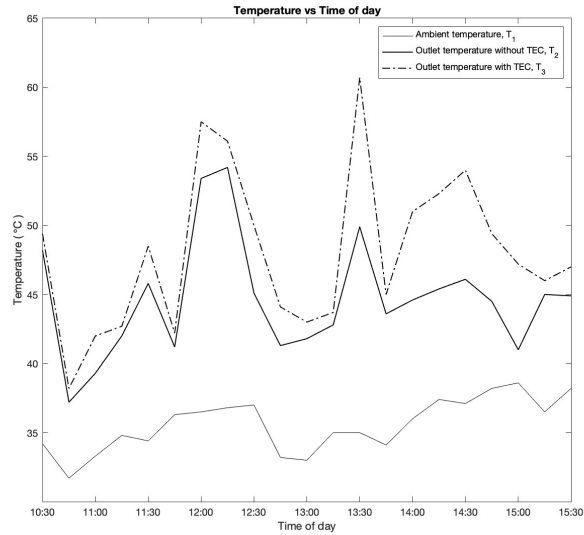


Figure 4.8: Variation of temperature with time of day

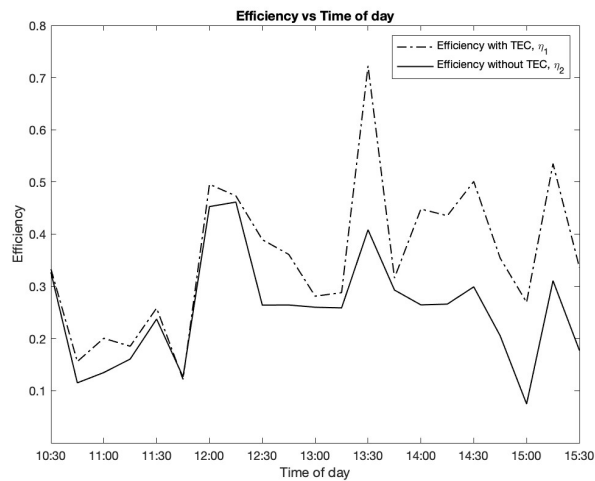


Figure 4.9: Variation of instantaneous efficiency with time of day

4.4 Performance Analysis III

Figure 4.10 shows the variation of solar intensity with the time of day for the third day of the experiment. The solar intensity increased steadily during the morning hours reaching a peak value at midday. After this point, the insolation started reducing gradually until evening. During the experiments, the highest value of solar intensity was measured as 362 W/m^2 at 12:30 PM with an average value of insolation throughout the day coming out to 210 W/m^2 . The voltage and current graphs shown in Fig 4.11 and Fig 4.12 followed the same trend as the insolation, increasing as sunlight increases and decreasing when the sun goes down.

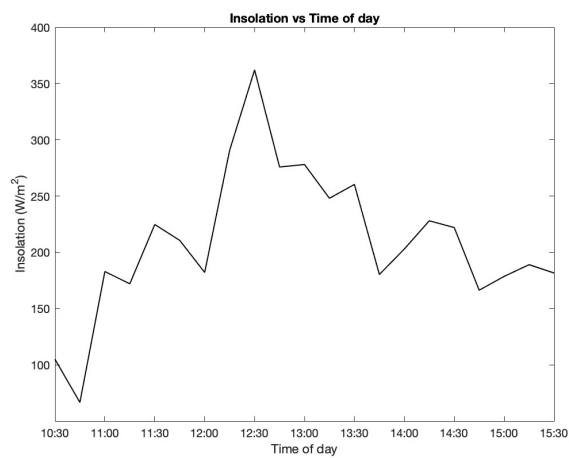


Figure 4.10: Variation of insolation with time of day

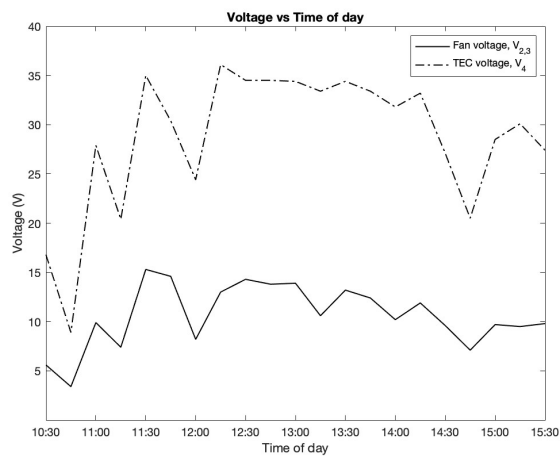


Figure 4.11: Variation of voltage across fans and TECs with time of day

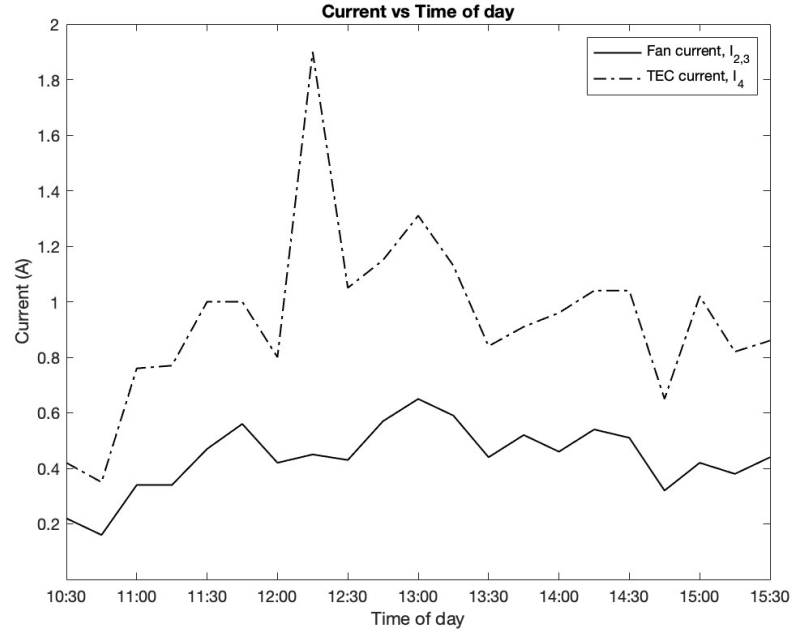


Figure 4.12: Variation of current across fans and TECs with time of day

4.4.1 Temperature and Efficiency Analysis III

On the third day of the experiment, Figure 4.13 depicts how the temperature changed with the time of day. The air temperature emanating from the SAHs was found to be generally rising from the morning into the afternoon. The greatest temperature recorded at 1:15 PM for the modified and conventional SAHs was 45.2°C and 43.6°C, respectively. After that, the temperature began to drop gradually until nightfall. The outside temperature rose until 11:30 AM and then momentarily dropped. The average recorded ambient temperature was 33°C, while the highest recorded figure was 35.7°C. The outlet temperature of the modified SAH was consistently higher than that of the traditional SAH by around 2-3°C, indicating a greater heat gain.

Figure 4.14 shows the instantaneous efficiencies at different times of the day. The maximum efficiency reached by the conventional SAH was 30% and 35% for the modified SAH at 12:15 PM midday offering an efficiency improvement of about 17%.

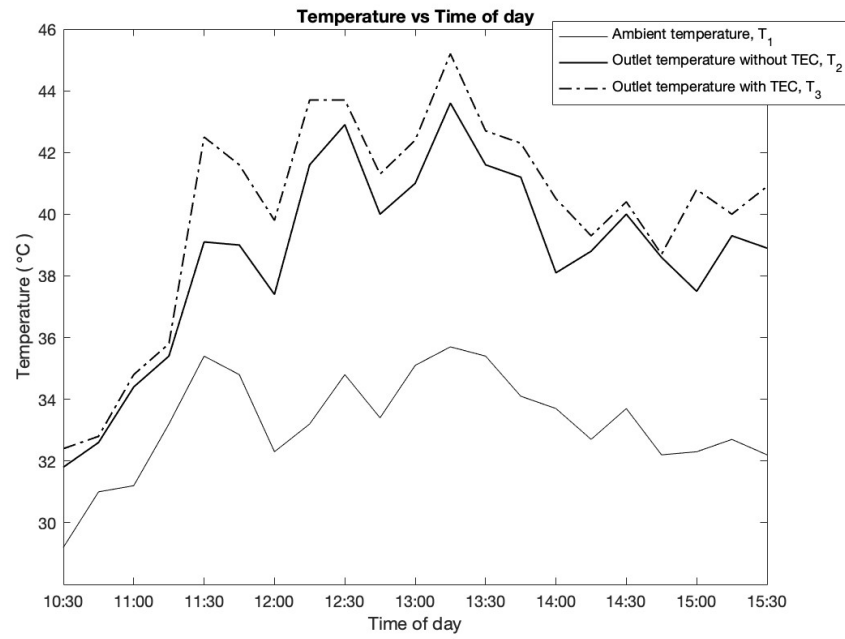


Figure 4.13: Variation of temperature with time of day

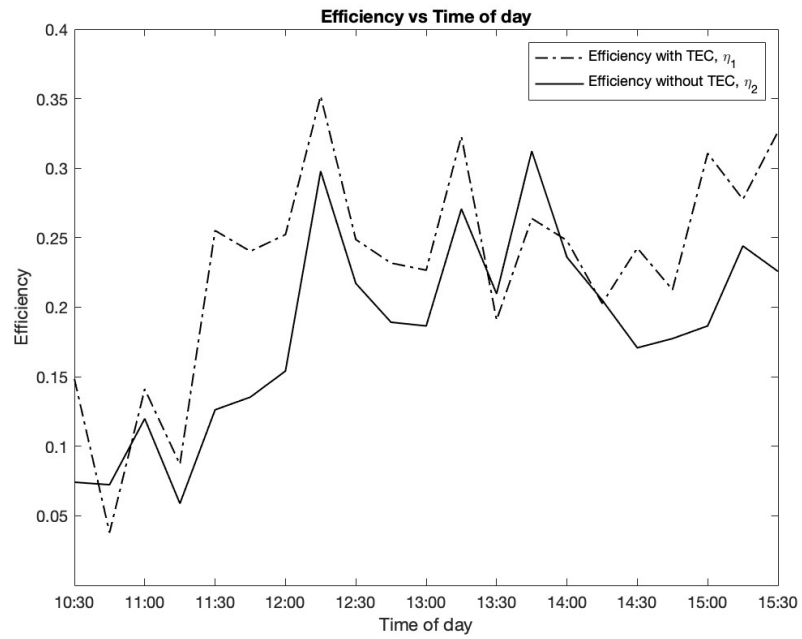


Figure 4.14: Variation of instantaneous efficiency with time of day

Chapter 5

Conclusions and Recommendations

5.1 Conclusions

Figure 4.13 shows how the temperature varied with the time of day on the third day of the experiment. It was discovered that, on average, the air temperature coming from the SAHs was increasing from the morning into the afternoon. For the modified and conventional SAHs, the highest temperature was 45.2°C and 43.6°C, respectively, recorded at 1:15 PM. Then the temperature started to steadily decline till dusk. The temperature outside increased until 11:30 AM, at which point it briefly decreased. The highest recorded temperature was 35.7°C, while the average recorded ambient temperature was 33°C. The modified SAH continuously had an outlet temperature that was 2 – 3°C higher than the standard SAH, indicating a stronger heat gain. The exit air temperature of the solar air heater with TECs mounted to its absorber plate increased noticeably. The air enters both collectors at the same temperature and flows through them at roughly the same rate, but the exit temperatures varied after each subsequent reading, with the solar air heater with TECs registering a higher temperature.

The thermoelectric coolers improved the solar air heater by continuously absorbing heat from the absorber plate and transferring it to the flowing air by conduction at a pace significantly faster than the absorber without TECs.

The system's heat loss was caused by various elements, including the glazing's ability to reflect incident radiation and conduction losses from the collector's sides and other system components. These losses also lowered the system's efficiency.

5.2 Recommendations

The following suggestions are made in light of the enhanced performance of the solar air heater provided by the thermoelectric coolers attached:

You must have the necessary understanding of the setup and operation of the exper-

iment. To achieve their best performance, the TECs should be linked so that they utilize their maximum current rating. To do this, the series connection had been set up. Similarly, as the experiment's solar panels were insufficient to provide the current required for the TECs to operate at peak efficiency, more solar panels are required to ensure that the TECs operate at their maximum rating.

Furthermore, to guarantee consistent solar insolation, this experiment should be conducted during the dry season. This is because the experiment's results were impacted by the sun's continuous variations in insolation.

Moreover, human error and inaccurate measuring techniques also impacted the data obtained from this experiment; for this reason, digital instruments that can be programmed to take measurements automatically at different intervals are advised. Additionally, it will guarantee that all of the measured parameters' readings are obtained at the same solar insolation.

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Appendix A

Table of Results

A.1 day 3 of the experiment

Table A.1: values obtained for experiment 3

I	I_2	V_2	I_3	V_3	I_4	V_4	T_1	T_2	T_3	Ve_{1i}	Ve_{1ii}	Ve_{2i}	Ve_{2ii}
105	0.22	5.3	0.22	5.6	0.42	16.8	29.2	31.8	32.4	2.2	1.6	1.2	1
66.6	0.16	3.1	0.17	3.4	0.35	8.9	31	32.6	32.2	1.1	0.5	1.3	0.9
182.9	0.34	10	0.35	9.9	0.76	27.9	31.2	34.4	34.8	3.5	2.4	2.9	2.2
172	0.34	6.5	0.35	7.4	0.77	20.4	33.2	35.4	35.8	2.6	2	1.7	1.7
224.7	0.47	14.9	0.55	15.3	1	35	35.4	39.1	42.5	4	3	3.1	2.8
210.6	0.56	12	0.66	14.6	1	30.4	34.8	39	41.6	3.9	2.5	2.9	2.4
182.1	0.42	7.9	0.43	8.2	0.8	24.4	32.3	37.4	39.8	2.9	2.1	2.1	2
290.6	0.45	11.7	0.47	13	1.9	36.1	33.2	41.6	43.7	5.5	3.3	4.1	3.6
362.1	0.43	14.7	0.43	14.3	1.05	34.5	34.8	42.9	43.7	5	3.2	4	3.2
275.8	0.57	12.9	0.59	13.8	1.15	34.5	33.4	40	41.3	4.3	2.6	3.3	2.7
278	0.65	12.6	0.64	13.9	1.31	34.4	35.1	41	42.4	4.4	3.1	3.8	2.9
248	0.59	9.8	0.59	10.6	1.13	33.4	35.7	43.6	45.2	4.4	2.8	3.4	3
260.3	0.44	12.6	0.45	13.2	0.84	34.4	35.4	41.6	42.7	3.2	2.4	3.5	3.1
180.3	0.52	11.7	0.54	12.4	0.91	33.4	34.1	41.2	42.3	2.6	2.5	3.6	2.5
202.9	0.46	9.9	0.46	10.2	0.96	31.8	33.7	46.1	40.5	3.6	2.7	3	2.2
228	0.54	10.9	0.53	11.9	1.04	33.2	32.7	38.8	39.3	3.5	2.5	3.3	2.5
222	0.51	9.2	0.49	9.6	1.04	27.1	33.7	40	40.4	4	2.7	2.4	2.1
166.3	0.32	6.7	0.32	7.1	0.65	20.5	32.2	38.6	38.7	2.5	1.8	1.7	1.7
178.6	0.42	8.9	0.41	9.7	1.02	28.5	32.3	37.5	40.8	3.4	2.2	2.8	2
189	0.38	10	0.37	9.5	0.82	30.1	32.7	39.3	40	3.8	2.2	2.9	2.3
181.6	0.44	9	0.45	9.8	0.86	27.4	32.2	38.9	40.9	3.3	2.4	2.6	2

A.2 day 2 of the experiment

Table A.2: values obtained for experiment 2

I	I_2	V_2	I_3	V_3	I_4	V_4	T_1	T_2	T_3	Ve_{1i}	Ve_{1ii}	Ve_{2i}	Ve_{2ii}
262.1	0.75	6.7	0.75	8.4	1.39	34.6	34.2	48.2	49.4	3.1	1.9	2.6	2
277.5	0.69	13.9	0.75	15.3	1.4	35.2	31.7	37.2	38.2	3.7	2.2	2.5	2
311.2	0.7	13.5	0.81	15.1	1.48	36.6	33.3	39.3	42	3.8	2.5	2.8	2.6
321.8	0.78	15.8	0.87	16.6	1.61	38.4	34.8	42	42.7	4	2.8	3	2.6
346.7	0.88	15.8	0.89	16.4	1.7	37.2	34.4	45.8	48.5	3.5	2.2	3.2	2.4
332.5	0.88	15.7	0.89	16.8	1.7	36.9	36.3	41.2	42.2	3.6	2.6	3.5	3.2
380.6	0.98	16.2	1.05	17.5	1.91	40.3	36.5	53.4	57.5	5.3	2.9	4.4	3.6
375.9	1.05	15.1	1.03	17.1	1.97	44.1	36.8	54.2	56.1	5.6	3	4.4	3.4
228.1	0.57	10.2	0.55	10.5	1.16	31.8	37	45.1	50	3.7	2.2	3	2.6
204.4	0.47	9.8	0.46	10.1	0.91	30.5	33.2	41.3	44.1	3.6	2.1	2.9	2.1
243	0.58	11.2	0.57	11.6	1.3	33.9	33	41.8	43	3.8	2.2	4	3.1
205.5	0.44	9.3	0.44	10.1	0.93	30	35	42.8	43.7	3.5	2.2	2.6	2.5
342.2	0.88	14.8	0.89	16.9	1.62	41.7	35	49.9	60.7	5.7	3	3.9	3.4
327.8	0.89	16	0.88	17.2	1.46	35	34.1	43.6	45	5.4	3	4.6	3.3
315.7	0.83	15.7	0.88	16.9	1.6	36.7	36	44.6	51	5.7	2.8	4.4	3.2
303.1	0.83	15.6	0.88	16.5	1.61	35.1	37.4	45.4	52.3	5	3	4.4	3.5
301.2	0.83	15.8	0.84	16	1.62	35.4	37.1	46.1	54	5.5	2.6	4.1	3.7
285.8	0.8	16.2	0.83	16.5	1.55	34.5	38.2	44.5	49.4	5.2	3	3.8	3.5
312.2	0.75	15	0.76	16.1	1.45	35.4	38.6	41	47.2	5.6	3	4	3.5
78.3	0.18	4.1	0.19	4.5	0.42	17	36.5	45	46	2.2	1.3	1.3	0.8
87.4	0.2	4.4	0.22	4.9	0.45	13.5	38.2	44.9	47	1.6	1	0.9	0.8

A.3 day 1 of the experiment

Table A.3: values obtained for experiment 1

I	I_2	V_2	I_3	V_3	I_4	V_4	T_1	T_2	T_3	Ve_{1i}	Ve_{1ii}	Ve_{2i}	Ve_{2ii}
262.1	0.75	6.7	0.75	8.4	1.39	34.6	34.2	48.2	49.4	3.1	1.9	2.6	2
277.5	0.69	13.9	0.75	15.3	1.4	35.2	31.7	37.2	38.2	3.7	2.2	2.5	2
311.2	0.7	13.5	0.81	15.1	1.48	36.6	33.3	39.3	42	3.8	2.5	2.8	2.6
321.8	0.78	15.8	0.87	16.6	1.61	38.4	34.8	42	42.7	4	2.8	3	2.6
346.7	0.88	15.8	0.89	16.4	1.7	37.2	34.4	45.8	48.5	3.5	2.2	3.2	2.4
332.5	0.88	15.7	0.89	16.8	1.7	36.9	36.3	41.2	42.2	3.6	2.6	3.5	3.2
380.6	0.98	16.2	1.05	17.5	1.91	40.3	36.5	53.4	57.5	5.3	2.9	4.4	3.6
375.9	1.05	15.1	1.03	17.1	1.97	44.1	36.8	54.2	56.1	5.6	3	4.4	3.4
228.1	0.57	10.2	0.55	10.5	1.16	31.8	37	45.1	50	3.7	2.2	3	2.6
204.4	0.47	9.8	0.46	10.1	0.91	30.5	33.2	41.3	44.1	3.6	2.1	2.9	2.1
243	0.58	11.2	0.57	11.6	1.3	33.9	33	41.8	43	3.8	2.2	4	3.1
205.5	0.44	9.3	0.44	10.1	0.93	30	35	42.8	43.7	3.5	2.2	2.6	2.5
342.2	0.88	14.8	0.89	16.9	1.62	41.7	35	49.9	60.7	5.7	3	3.9	3.4
327.8	0.89	16	0.88	17.2	1.46	35	34.1	43.6	45	5.4	3	4.6	3.3
315.7	0.83	15.7	0.88	16.9	1.6	36.7	36	44.6	51	5.7	2.8	4.4	3.2
303.1	0.83	15.6	0.88	16.5	1.61	35.1	37.4	45.4	52.3	5	3	4.4	3.5
301.2	0.83	15.8	0.84	16	1.62	35.4	37.1	46.1	54	5.5	2.6	4.1	3.7
285.8	0.8	16.2	0.83	16.5	1.55	34.5	38.2	44.5	49.4	5.2	3	3.8	3.5
312.2	0.75	15	0.76	16.1	1.45	35.4	38.6	41	47.2	5.6	3	4	3.5
78.3	0.18	4.1	0.19	4.5	0.42	17	36.5	45	46	2.2	1.3	1.3	0.8
87.4	0.2	4.4	0.22	4.9	0.45	13.5	38.2	44.9	47	1.6	1	0.9	0.8